

DANISH ARSHID

AI/ML Engineer | Python Developer | LLM & RAG / Agentic Systems

danisharshidmantoo@gmail.com | +91-9906827451 | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | NIT Srinagar, CSE '26

EDUCATION

National Institute of Technology, Srinagar

2022 - 2026

B.Tech in Computer Science & Engineering | CGPA: 8.11/10

EXPERIENCE

AI/ML Engineer Intern — Robonox

Dec 2024 - Feb 2025

Remote | TechFusion 2.0, 45-Day Internship & Training Program

- Designed and deployed a real-time computer vision pipeline in Python using TensorFlow and OpenCV, achieving 95% accuracy at 20+ FPS on live video streams.
- Reduced inference latency by ~30% through optimized preprocessing and pipeline profiling, improving real-time throughput and applying core data structures & algorithms (DSA) for efficient processing.
- Managed the full ML workflow — preprocessing, training, evaluation, and deployment — in a Linux environment using production-style development practices.

PROJECTS

Digital Twin — Agentic AI Career Assistant

2026

Python, Gradio, Groq/OpenAI SDK, OpenRouter, Render

- Built and deployed a tool-calling AI agent that resolves multi-step function-calling loops using the OpenAI SDK before returning a final response.
- Implemented an LLM-as-judge guardrail with a Pydantic-constrained structured output (binary classification) to filter off-topic responses, with automatic retry and corrective re-prompting on failure.
- Orchestrated multiple LLM providers (Groq for the primary agent, OpenRouter for evaluation) and deployed the full application end-to-end on Render.

Multi-Agent Research Assistant

2025

Python, FastAPI, LLM APIs, RAG, Docker

- Developed a multi-agent AI research assistant with an Orchestrator Agent that decomposes complex user queries and coordinates specialized Research, Analysis, and Summary agents for multi-step reasoning using LLMs.
- Built a Retrieval-Augmented Generation (RAG) pipeline to ground agent responses in retrieved context, improving factual accuracy of LLM outputs.
- Designed an abstraction layer decoupling agent logic from any single LLM provider, enabling swappable model backends without core logic changes.

Resume Evaluation & Candidate Management System

2025

React, FastAPI, Python, MongoDB, LLM APIs, RAG

- Built a full-stack system that extracts structured data from resumes (PDF parsing) and job descriptions, then feeds both into an LLM-based RAG pipeline for automated candidate evaluation.
- Implemented AI-generated match scoring, compatibility analysis, and improvement recommendations comparing candidates against job requirements using Python and DSA-driven ranking logic.
- Persisted evaluation results in MongoDB and built a React frontend for reviewing AI-generated insights and rankings.

ML Foundations From Scratch — Autograd Engine & Character-Level Language Model

2025

Python, NumPy, PyTorch, Graphviz

- Implemented a reverse-mode automatic differentiation engine from scratch (computational graphs, operator overloading, chain-rule gradients), then built a Neuron → Layer → MLP stack on top and trained it end-to-end with gradient descent, without PyTorch/TensorFlow; added a Graphviz visualizer for gradient flow.
- Built a character-level bigram language model (27×27 transition matrix, negative log-likelihood), then reimplemented it as a neural network with one-hot encoding, softmax, backpropagation, and L2 regularization.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, SQL

AI/ML: LLM APIs, RAG, Agentic / Tool-Calling Systems, LangChain, LlamaIndex, Prompt Engineering, LLM-as-Judge Evaluation, Multi-Agent Systems, TensorFlow, PyTorch, Scikit-learn, OpenCV

Core CS: Data Structures & Algorithms (DSA), OOP, REST APIs, JWT Authentication, RBAC, Secure API Design

Backend & Frameworks: Node.js, Express.js, FastAPI, Java Spring Boot, Gradio

Frontend: React.js, Bootstrap, HTML, CSS

Databases: MongoDB, MySQL, PostgreSQL

Cloud & Tools: Docker, Linux/Unix, Git, GitHub, Render, Groq, OpenRouter, Jupyter Notebook, Postman

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI & Stanford Online (Andrew Ng), via Coursera
- 100 Days of Code: The Complete Python Pro Bootcamp — Angela Yu (Udemy)
- TechFusion 2.0 — AI/ML & Deep Learning Internship, Robonox
- Achievement: Solved 600+ DSA problems on LeetCode and GeeksforGeeks, building a strong foundation in algorithms, data structures, and OOP.